Capstone Challenge 2024-2025

Grade 10 – semester 2

Harvesting and storing energy from unconventional resources

Egyptian Grand Challenge:

- Improve the use of alternative energies.
- Improve the scientific and technological environment for all.
- Address & reduce pollution in air, water and soil.
- Reduce and adapt to the effect of the climatic change.

(*The team can add any more Egyptian grand challenges according to the chosen method of their idea*)

Semester theme:

Energy, Force and Power

Big Idea:

Energy is essential for human development. Usually, energy is produced from sources that result in environmental deficiencies, which is not desirable. Moreover, current energy storing systems suffer from serious drawbacks. Thus, design a simple, measurable and cost-effective system that can generate electricity from unconventional resources and store it effectively.

Essential question:

How can you utilize sustainable resources to harvest and store clean energy?

Capstone design challenge:

The rapidly growing population together with the expected depletion of traditional energy sources driven from fossil fuels have motivated the design of alternative solutions to produce energy from non-conventional sustainable resources. Students must use one of the following technologies to produce and store energy:

- Micro sized hydro power station,
- Small solar chimney,

- Innovative vertical wind axis turbine,
- Biofuel,
- Microbiological cells,
- Liquid fuel cells,
- Pumped hydro storage (PHS).

In addition, the uncertainty and climatological dependency of most renewable energy sources mandates the application of energy storage systems that work in a time of deficiency. Therefore, the students must implement a simple and measurable storage system for the electricity they produced.

Your device must meet the following:

- Produce at least 0.04 kWh in at most 5 minutes.
- Store the energy in any storage device to be used anytime.
- Using waste material in your construction is preferable.
- Prototypes should not exceed 0.5 m^3 in volume.

Design requirement:

Choose a proper design requirement(s) based on your idea, for example:

- User friendly.
- Environmentally friendly.
- Fast in producing energy.
- Non-impediment (your device will not negatively affect the students' activities).

You may find another design requirement (s) according to your idea.

Materials:

- You are allowed to use some laboratory equipment to test your device.
- The used material is preferred to be eco-friendly.

Constraints:

- Use ONLY fab lab tools to create your device, no external fabricated organizations are allowed.
- You must follow All School Safety rules and documents in your Capstone. Failure to do so will result in a reduced grade.
- No equipment or materials that belong to the laboratories can be removed from the laboratories at any time. You must conduct those tests that require this equipment in the labs. Failure to do so will result in a reduced grade.
- The production of electricity from piezoelectric effect is not permitted.
- Storing energy in ready-made power banks is not accepted.
- Hydrogen is not allowed to be used or included in the prototype.
- In the case of storing energy in the form of compressed fluids, pressure should not exceed 2 bar.

Mandatories:

- On the exhibition day, students are required to show the portfolio and documentation of the real tests (videos, reports, analysis, etc.) they performed.
- On the exhibition day, they must also prove for evaluators that their prototypes are workable, testable and measurable.
- You must secure all measuring tools on the exhibition day.